

Question Paper

Exam Date & Time: 01-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

SEVENTH SEMESTER B.TECH. DEGREE

MAKEUP EXAMINATIONS - JANUARY 2026

Deep learning [NPT_4124]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Evaluate the effects of model complexity and dataset size on overfitting and underfitting. How can cross-validation help mitigate these issues? (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Compare and contrast supervised and unsupervised learning algorithms. Identify scenarios where one is more suitable than the other. (5)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Examine the role of non-linearity in activation functions and its importance in enabling deep networks to approximate complex mappings. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Compare the performance and convergence characteristics of sigmoid, ReLU, and ELU in deep neural architectures. (5)
- Course Outcome: "CO4" / Cognitive Level: "Evaluate"
- 3) Examine how modern activation functions such as Leaky ReLU, GELU, or Swish improve learning stability compared to ReLU. (5)
- A) Course Outcome: "CO4" / Cognitive Level: "Analysis"
- B) Analyze the difference between early stopping and regularization in terms of their effect on model performance. In which situations would one be preferred over the other? (5)
- Course Outcome: "CO4" / Cognitive Level: "Analysis"
- 4) Evaluate the limitations of standard RNNs in modeling long-term dependencies and explore methods like gradient clipping or truncated BPTT to address them. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Evaluate"
- B) Compare the effectiveness of bidirectional RNNs and GRUs for handling sequential data with contextual dependencies. (5)
- Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- 5) Evaluate the impact of vanishing and exploding gradients in RNNs. How do these issues affect the network's ability to learn long-term patterns? (5)

A)

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

- B) Evaluate the significance of the input, forget, and output gates in LSTMs. How do they contribute to the network's ability to remember long-term dependencies? (5)

Course Outcome: "CO1" / Cognitive Level: "Evaluate"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 NPT_4124:Deep learning

#	Course Outcome	Description	Marks
1	1	CO 1	15
2	2	CO 2	15
3	4	CO 4	20
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 NPT_4124:Deep learning

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	0	0(0%)
4	Analysis	5	25(50%)
5	Create	5	25(50%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)